

E066

Object Oriented Programming in Python

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A fleet app tracks 300 delivery trucks for a regional grocer. Each Truck needs to start and stop its engine, and an Engine class already has start() and stop(). The grocer plans to add electric trucks next year. How should Truck reuse Engine?

- A. Composition: give Truck an engine attribute that holds an Engine object.
- B. Make Truck inherit from Engine.
- C. Always use inheritance whenever code can be reused.
- D. Make Engine inherit from Truck.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

Which relationship is the best fit for inheritance?

- A. A Library has many Books.
- B. A Car has an Engine.
- C. An Order uses a PaymentService.
- D. A SavingsAccount is a kind of BankAccount.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

An HR system tracks 450 employees. class Employee sets self.name in __init__(self, name). class Manager(Employee) defines __init__(self, name, reports) and sets only self.reports. Calling Manager('Ana', 4).name raises AttributeError. What is the fix?

- A. Call super().__init__(name) inside Manager.__init__.
- B. Nothing; Python runs Employee.__init__ automatically.
- C. Make Manager inherit from Employee twice.
- D. Reduce the number of employees tracked.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A drawing app written by a team in two countries has a Shape base class whose area() returns 0. Circle and Square override area(). The code runs total = sum(s.area() for s in [Circle(1), Square(2)]). What is total, to two decimals?

- A. 7.14
- B. An error, because the list mixes types.
- C. 4
- D. 0

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A credit union's BankAccount class has been in use since 2018. Its balance must never be set below zero by outside code. Which design best protects it?

- A. Make balance a class attribute.
- B. Keep balance public and write a comment asking callers not to go negative.
- C. Store it as _balance and expose a @property whose setter rejects negative values.
- D. Name it __balance, which makes it impossible to reach from outside.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

class Payment(ABC) declares process() and refund() as @abstractmethod. The payments team of five writes class CardPayment(Payment) that defines only refund(). What happens on CardPayment()?

- A. It works, because refund() is implemented.
- B. TypeError: the class cannot be instantiated.
- C. It works; an error appears only when process() is called.
- D. Payment() works but CardPayment() does not.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A sports league app with 32 teams defines class Team: members = [] and def add(self, m): self.members.append(m). After a = Team(); b = Team(); a.add('Kai'), the league's logo loads from a file. What does print(b.members) show, and why?

- A. It depends on whether the logo file loaded.
- B. An error, because b never called add().
- C. [], because each object has its own list.
- D. ['Kai'], because both objects share the class-level list; set self.members = [] in __init__.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

class Counter: count = 0. Then c1 = Counter(); c2 = Counter(); c1.count = 5. What does print(Counter.count, c2.count) show?

- A. 5 5
- B. 0 0
- C. 0 5
- D. 5 0

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A text editor used by 20,000 writers needs an undo feature. The most recent change must be undone first. The editor is written in Python 3.11. Which structure fits?

- A. A set of changes.
- B. A queue.
- C. A stack (list with append and pop).
- D. A sorted list of changes by size.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

A print server must handle jobs in the order they arrive. Which structure and operations fit?

- A. A heap ordered by page count.
- B. collections.deque with append and popleft.
- C. A set of job IDs.
- D. A list with append and pop().

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A tutoring site with 1,500 students uses `def total(n): return 0 if n == 0 else n + total(n - 1)`. The site's servers run on Linux. What does `total(4)` return?

- A.** `RecursionError`
- B.** 24
- C.** 10
- D.** 4

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

`fact(n)` returns `n * fact(n - 1)`. Which base case makes `fact(3)` return 6?

- A.** if `n == 0`: return 0
- B.** No base case; Python stops at 0 automatically.
- C.** if `n == 0`: return `n * fact(n - 1)`
- D.** if `n == 0`: return 1

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A fleet app ~~tracks 300 delivery trucks for a regional grocer~~. Each Truck needs to start and stop its engine, and an Engine class already has start() and stop(). ~~The grocer plans to add electric trucks next year~~. How should Truck reuse Engine?

Target word: choose the relationship **Decides:** A truck has an engine; it is not a kind of engine.

A	Correct	Has-a relationship; Truck delegates to its engine.
B	Confusion set	A truck is not a kind of engine.
C	Absolute language	Reuse alone does not justify inheritance.
D	Inverted operation	Reverses the relationship and is still not is-a.

2. Answer D Concept 1 • transfer

Which relationship is the best fit for inheritance?

Target word: find the true is-a **Decides:** Only one pair is “X is a kind of Y.”

A	Inverted operation	Has-a (a collection), which is composition.
B	Confusion set	Has-a, which is composition.
C	Adjacent category	A uses/depends-on relationship, not is-a.
D	Correct	True is-a: inheritance fits.

3. Answer A Concept 2

~~An HR system tracks 450 employees~~. class Employee sets self.name in __init__(self, name). class Manager(Employee) defines __init__(self, name, reports) and sets only self.reports. Calling Manager('Ana', 4).name raises AttributeError. What is the fix?

Target word: why the parent attribute is missing **Decides:** The child's __init__ never runs the parent's.

A	Correct	Runs Employee's setup so self.name exists.
B	Confusion set	An overriding __init__ replaces the parent's unless you call it.
C	Adjacent category	Not valid and does not run the constructor.
D	Camouflage	Data size has nothing to do with the missing attribute.

4. Answer A Concept 3

A drawing app ~~written by a team in two countries~~ has a Shape base class whose area() returns 0. Circle and Square override area(). The code runs total = sum(s.area() for s in [Circle(1), Square(2)]). What is total, to two decimals?

Target word: result of a polymorphic call **Decides:** Each object runs its own area().

A	Correct	$\pi \cdot 1^2 \approx 3.14$ plus $2^2 = 4$.
B	Confusion set	Python lists can mix types; each object just answers area().
C	Dropped element	Counts only the square.
D	Inverted operation	Assumes the base-class method runs instead of the overrides.

5. Answer C Concept 4

A credit union's BankAccount class ~~has been in use since 2018~~. Its balance must never be set below zero by outside code. Which design best protects it?

Target word: protect state with validation **Decides:** Changes must pass a check.

A	Adjacent category	Shares one balance across all accounts and still does not validate.
B	Dropped element	No enforcement.
C	Correct	Controls every change through one validated path.
D	Absolute language	Name mangling hides it but is still reachable; it also does no validation.

6. Answer B Concept 5

class Payment(ABC) declares process() and refund() as @abstractmethod. ~~The payments team of five writes~~ class CardPayment(Payment) that defines only refund(). What happens on CardPayment()?

Target word: effect of a missing abstract method **Decides:** One abstract method is still unimplemented.

A	Dropped element	Ignores process().
B	Correct	Every abstract method must be implemented first.
C	Confusion set	That is the raise-NotImplementedError pattern, not ABC.
D	Inverted operation	The abstract parent itself cannot be instantiated.

7. Answer D Concept 6

A sports league app with 32 teams defines class Team: members = [] and def add(self, m): self.members.append(m). After a = Team(); b = Team(); a.add('Kai'), ~~the league's logo loads from a file~~. What does print(b.members) show, and why?

Target word: shared mutable class attribute **Decides:** members is one list owned by the class.

A	Camouflage	The logo does not affect attributes.
B	Dropped element	b can still read the class attribute.
C	Confusion set	That would be true only if the list were created in __init__.
D	Correct	append changes the shared list.

8. Answer B Concept 6 · transfer

class Counter: count = 0. Then c1 = Counter(); c2 = Counter(); c1.count = 5. What does print(Counter.count, c2.count) show?

Target word: effect of assigning through an instance **Decides:** Assigning c1.count creates an instance attribute on c1 only.

A	Confusion set	Assumes assignment changes the shared class attribute.
B	Correct	The class attribute is unchanged; c2 still reads it.
C	Inverted operation	Puts the change on the wrong object.
D	Adjacent category	Would require changing Counter.count directly.

9. Answer C Concept 7

A text editor used by 20,000 writers needs an undo feature. The most recent change must be undone first. The editor is written in Python 3.11. Which structure fits?

Target word: structure for undo **Decides:** Most recent first = last-in, first-out.

A	Dropped element	Sets have no order at all.
B	Confusion set	FIFO would undo the oldest change first.
C	Correct	LIFO returns the latest change first.
D	Adjacent category	Size order is not time order.

10. Answer B Concept 7 · transfer

A print server must handle jobs in the order they arrive. Which structure and operations fit?

Target word: FIFO structure **Decides:** Arrival order = first-in, first-out.

A	Confusion set	A priority queue, not arrival order.
B	Correct	Efficient FIFO queue.
C	Adjacent category	Unordered.
D	Inverted operation	pop() takes the newest job: LIFO.

11. Answer C Concept 8

A tutoring site with 1,500 students uses `def total(n): return 0 if n == 0 else n + total(n - 1)`. The site's servers run on Linux. What does `total(4)` return?

Target word: trace the recursion **Decides:** $4 + 3 + 2 + 1 + 0$.

A	Adjacent category	Happens only without a reachable base case; this one has one.
B	Confusion set	$4!$ multiplies instead of adding.
C	Correct	Sums 4 down to the base case 0.
D	Dropped element	Stops after the first call.

12. Answer D Concept 8 · transfer

`fact(n)` returns $n * \text{fact}(n - 1)$. Which base case makes `fact(3)` return 6?

Target word: choose the base case **Decides:** The base value is multiplied into every result.

A	Inverted operation	Multiplying by 0 makes every result 0.
B	Absolute language	Python raises <code>RecursionError</code> instead.
C	Dropped element	Still recurses, so there is no real base case.
D	Correct	$3 \times 2 \times 1 \times 1 = 6$.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	7	
Wrong stem number	A figure copied from the prompt that is not the answer.	0	
Adjacent category	The right idea in the wrong bucket.	7	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	3	
Confusion set	Every option is real; the test is exact role-matching.	10	
Camouflage	Answers a stem fact that was never the question.	2	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Inheritance (is-a) vs. composition (has-a)	Inherit only when the child truly is a kind of the parent; when one object uses or contains another, hold it as an attribute.	Inheriting just to reuse a method.	Can you say “X is a Y” or only “X has a Y”?
2	Overriding <code>__init__</code> and <code>super()</code>	A child that overrides <code>__init__</code> must call <code>super().__init__(...)</code> or the parent’s setup never runs.	Assuming Python runs the parent constructor automatically.	<code>AttributeError</code> on a parent attribute in a subclass object.
3	Polymorphism	Calling the same method on objects of different classes runs each object’s own version.	Thinking a list of mixed types needs type checks, or that the base method runs.	A loop calling one method over a mixed collection.
4	Encapsulation with properties	Keep state in a “private” attribute (<code>_name</code>) and expose it through a <code>@property</code> whose setter validates changes.	Believing double underscores make data truly inaccessible, or leaving it public and documenting a rule.	A value that must never become invalid.
5	Abstract base classes	A class with an unimplemented <code>@abstractmethod</code> cannot be instantiated; a subclass must implement every abstract method first.	Expecting the error only when the missing method is called.	ABC, <code>@abstractmethod</code> , and a subclass missing one method.
6	Class vs. instance attributes	Class attributes are shared by all instances; assigning <code>self.x</code> creates an instance attribute. Mutable defaults belong in <code>__init__</code> .	Defining a list at class level and expecting each object to get its own.	One object’s change shows up in another.
7	Stack vs. queue	A stack is last-in, first-out (undo, call stack); a queue is first-in, first-out (job lines).	Using a queue for undo or a stack for fair ordering.	“Most recent first” vs. “in the order they arrived.”
8	Recursion and the base case	A recursive function needs a base case that returns without recursing, and each call must move toward it.	A wrong base value that poisons every result, or none at all.	A function that calls itself.

Course scope (public description): This course bridges foundational programming skills to advanced object-oriented concepts, empowering learners to design and implement robust, scalable applications. It emphasizes the core pillars of object-oriented programming such as inheritance, composition, encapsulation, abstraction, and polymorphism while introducing data structure fundamentals. Learners will define and extend class hierarchies, apply method overriding, work with polymorphic collections, and traverse data structures using iterative and recursive techniques. Through practical problem-solving, the course reinforces code reusability and equips learners to apply object-oriented principles for efficient, maintainable solutions.